

Software Requirements Specification

Quran Learning Platform

Online Educational System for Quran Schools, Teachers, and Students

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1. Introduction

This Software Requirements Specification (SRS) document provides a comprehensive description of the Quran Learning Platform, an online educational system designed for Quran schools, teachers, and students. The platform facilitates Quran memorization (Hifz), Tajweed instruction, examination administration, progress tracking, live virtual classes, and Islamic educational content delivery.

This document is intended for the development team, project stakeholders, quality assurance engineers, and system architects. It serves as the authoritative reference for all functional and nonfunctional requirements, external interfaces, and system constraints.

1.1 Purpose

The purpose of this SRS is to define the complete set of requirements for the Quran Learning Platform. It establishes the functional behavior, performance expectations, security requirements, and technical constraints that shall govern the design, implementation, and validation of the system.

This document shall be used as the basis for: software design and architecture decisions; development task planning and estimation; test case design and acceptance criteria; and contract agreements between the client and development vendor.

1.2 Scope

The Quran Learning Platform is a web-based educational management system with planned mobile application support. The system shall provide the following core capabilities:

- ☐ User authentication and role-based access control for Administrators, Teachers, and Students
- ☐ Student enrollment, profile management, and progress tracking
- ☐ Daily memorization (Hifz) task assignment and revision scheduling
- ☐ Examination creation, administration, and automated grading
- ☐ Audio-based memorization submission and teacher assessment
- ☐ Live virtual classroom sessions for memorization and Tajweed
- ☐ Tajweed lesson content with multimedia resources
- ☐ Islamic educational content including supplications, Seerah, and companion stories
- ☐ Notification system for announcements, reminders, and private messages

- Comprehensive reporting and analytics dashboard

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
Hifz	The memorization of the entire Quran or portions thereof.
Tajweed	The set of rules for the correct pronunciation during Quranic recitation.
Surah	A chapter of the Quran; there are 114 Surahs in total.
Ayah	A verse of the Quran; the smallest division of Quranic text.
Juz	One of thirty parts into which the Quran is divided; also called Para.
Makharij	The points of articulation for Arabic letters in Tajweed.
Seerah	The biographical study of the life of Prophet Muhammad (PBUH).
RBAC	Role-Based Access Control — method of restricting access based on user roles.
JWT	JSON Web Token — compact, URL-safe means of representing claims.
MVP	Minimum Viable Product — version with sufficient features for early adopters.
FCM	Firestore Cloud Messaging — cross-platform messaging solution.
WebRTC	Web Real-Time Communication — API for browser-based real-time communication.
RTL	Right-to-Left — text direction for Arabic and similar scripts.
GDPR	General Data Protection Regulation — EU data protection law.

1.4 References

Reference	Description
[REF-1]	IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
[REF-2]	ISO/IEC/IEEE 29148:2018 — Systems and Software Engineering — Life Cycle Processes — Requirements Engineering.
[REF-3]	Project Specification Document — Quran Learning Platform, v1.0.
[REF-4]	OWASP Top 10 (2021) — Web Application Security Standards.
[REF-5]	WCAG 2.1 Guidelines — Web Content Accessibility Guidelines.
[REF-6]	GDPR Regulation (EU) 2016/679 — Data Protection Requirements.

1.5 Document Overview

The remainder of this document is organized as follows: Section 2 provides an overall description of the product, including user classes and operating environment. Section 3 details each system feature with functional requirements. Section 4 describes external interface requirements. Section 5 specifies nonfunctional requirements. Section 6 covers data requirements. Section 7 outlines the phased MVP roadmap. Appendix A documents the recommended technology stack.

2. Overall Description

2.1 Product Perspective

The Quran Learning Platform is a standalone web application with a planned native mobile companion. It operates as a centralized educational management hub connecting three primary stakeholder groups: school administrators who configure and oversee the platform, teachers who deliver instruction and assess student performance, and students who engage with learning materials and submit assignments.

The platform shall integrate with third-party services for real-time communication (live classes), push notifications, and cloud-based media storage. It maintains its own identity provider for authentication but may be extended to support OAuth 2.0 providers in future releases.

2.2 Product Functions

The following functional areas summarize the system's capabilities. Detailed requirements for each function are provided in Section 3.

Function Area	Description
Authentication	User registration, login, password management, and role-based access
Student Management	Profile management, class assignment, and progress monitoring
Memorization Tasks	Daily Quran memorization and revision task assignment and tracking
Examinations	Weekly/monthly exam creation, automated grading, and results
Notifications	Announcements, private messages, and automated alerts
Messaging	Direct teacher-student text communication
Memorization Assessment	Audio submission and multi-dimension teacher evaluation
Tajweed Lessons	Structured Tajweed instruction with multimedia resources
Live Classes	Real-time virtual classroom sessions
Islamic Content	Supplications, Seerah, and companion biographies
Reports & Analytics	Performance tracking and data visualization
Content Management	Administrative tools for educational content

2.3 User Classes and Characteristics

The system supports three primary user classes, each with distinct roles, privileges, and interaction patterns:

2.3.1 Administrator

The Administrator has full system access and is responsible for platform governance. Administrators are typically school principals, IT directors, or designated platform managers. They require comprehensive reporting capabilities and bulk operation tools. Key characteristics include:

- ☐ Full CRUD (Create, Read, Update, Delete) permissions on all user accounts
- ☐ Content management authority across all educational modules
- ☐ System configuration and platform setting control
- ☐ Access to aggregate statistics and analytics dashboards
- ☐ Announcement broadcasting to all user classes

2.3.2 Teacher

Teachers are the primary content creators and evaluators. They manage student cohorts, assign memorization tasks, conduct examinations, and provide formative feedback. Key characteristics include:

- ☐ Student management within assigned classes or groups
- ☐ Memorization task creation and assignment
- ☐ Examination authoring with multiple question types
- ☐ Audio submission review and Tajweed assessment
- ☐ Live class hosting and session management
- ☐ Direct messaging with students

2.3.3 Student

Students are the primary consumers of educational content. They access assigned tasks, submit memorization recordings, take examinations, and engage with Islamic educational materials. Key characteristics include:

- ☐ Personal dashboard with daily task overview
- ☐ Memorization audio recording and upload

- ☐ Examination participation
- ☐ Access to Tajweed lessons and Islamic content
- ☐ Live class attendance
- ☐ Messaging with assigned teachers

User Class	Primary Functions	Technical Proficiency
Administrator	Full system control	Intermediate to Advanced; comfortable with management interfaces
Teacher	Content creation, assessment	Basic to Intermediate; primary users of creation tools
Student	Content consumption, submission	Basic; intuitive interface required, ages 8+

2.4 Operating Environment

The system shall operate within the following technical environment:

- ☐ Server Infrastructure: Cloud-hosted (Render or Railway), Linux-based containers
- ☐ Database: PostgreSQL 15+ with full Unicode support for Arabic script
- ☐ Application Server: Node.js 20 LTS with Express.js framework
- ☐ Client Browsers: Chrome 110+, Firefox 120+, Safari 17+, Edge 120+
- ☐ Mobile: iOS 16+, Android 13+ (via planned React Native application)
- ☐ Network: HTTPS/TLS 1.3 required for all communications
- ☐ Media Storage: Cloud object storage for audio recordings and video content

2.5 Design and Implementation Constraints

The following constraints shall govern the system design and implementation:

- ☐ The backend API must be built using Node.js with Express.js and Drizzle ORM
- ☐ The frontend web application must use React with Tailwind CSS
- ☐ The database must be PostgreSQL with schema managed through Drizzle migrations
- ☐ Authentication must use JWT tokens with secure httpOnly cookie storage
- ☐ Push notifications must use Firebase Cloud Messaging (FCM)

- ☐ All Arabic text must render correctly with proper RTL support
- ☐ Audio recordings must support common formats: MP3, M4A, WAV (up to 20MB per file)
- ☐ The system must comply with GDPR and applicable data protection regulations

2.6 Assumptions and Dependencies

The following assumptions and dependencies are identified:

- ☐ Users have reliable internet connectivity (minimum 1 Mbps for audio, 5 Mbps for live video)
- ☐ Student devices have microphone capability for memorization recording
- ☐ Teachers have camera and microphone access for live class sessions
- ☐ Firebase Cloud Messaging service availability for push notifications
- ☐ Cloud storage service availability for media file hosting
- ☐ Correct Quranic text data (Uthmani script) will be provided or sourced from authenticated APIs

3. System Features

This section provides detailed functional requirements for each system feature. Each requirement is uniquely identified using the format [F-XXX] where F indicates a functional requirement and XXX is a sequential number. Priority levels are defined as: Critical (must have for MVP), High (important for initial release), Medium (desirable within first release), Low (future enhancement).

3.1 Authentication and Authorization

The Authentication and Authorization module provides secure identity verification, session management, and role-based access control for all system users.

3.1.1 User Registration

F-001 [Critical] The system shall allow Administrator users to create Teacher and Student accounts with email, password, full name, and role assignment.

F-002 [Critical] The system shall validate email uniqueness across all user accounts.

F-003 [Critical] Passwords shall be hashed using bcrypt with a minimum cost factor of 12 before storage.

F-004 [High] The system shall support self-registration for Students with teacher approval workflow.

F-005 [High] Registration shall include optional fields: phone number, date of birth, and profile photo.

3.1.2 User Login

F-006 [Critical] The system shall authenticate users via email and password.

F-007 [Critical] Upon successful authentication, the system shall issue a JWT access token (15-minute expiry) and a refresh token (7-day expiry).

F-008 [Critical] The system shall implement token refresh mechanism to maintain user sessions securely.

F-009 [High] The system shall implement rate limiting on login attempts (maximum 5 attempts per 15-minute window per IP).

F-010 [High] Failed login attempts shall be logged with timestamp, IP address, and user identifier for security auditing.

3.1.3 Role-Based Access Control

- F-011 [Critical]** The system shall enforce role-based access control (RBAC) with three roles: Administrator, Teacher, and Student.
- F-012 [Critical]** Administrators shall have unrestricted access to all system features and data.
- F-013 [Critical]** Teachers shall only access students assigned to their classes.
- F-014 [Critical]** Students shall only access their own data, assigned tasks, and permitted content.
- F-015 [High]** API endpoints shall validate user permissions on every request using middleware authorization checks.

3.1.4 Password Management

- F-016 [Critical]** Users shall be able to reset their password via email verification link (valid for 1 hour).
- F-017 [High]** Passwords must meet minimum complexity: 8+ characters, including uppercase, lowercase, and numeric character.
- F-018 [Medium]** The system shall maintain password history to prevent reuse of the last 5 passwords.

3.2 Student Management

The Student Management module enables teachers and administrators to manage student profiles, monitor academic progress, and review historical performance data.

3.2.1 Student Profiles

- F-019 [Critical]** The system shall store student profiles including: full name, email, enrollment date, assigned teacher, class/group, and current memorization level (Juz/Surah).
- F-020 [Critical]** Teachers and Administrators shall be able to view student profile details.
- F-021 [High]** Administrators shall be able to update student profile information and reassign students to different teachers.
- F-022 [High]** Students shall be able to view their own profile and update contact information.
- F-023 [Medium]** The system shall maintain a profile audit log tracking all changes with timestamp and modifying user.

3.2.2 Progress Monitoring

- F-024 [Critical]** Teachers shall be able to view a student's memorization progress chart showing completed Surahs, verses, and revision history.
- F-025 [High]** The progress view shall display memorization records with dates, assigned verses, and assessment scores.

F-026 [High] Teachers shall be able to view student examination results with filter by exam type (weekly/monthly) and date range.

F-027 [Medium] The system shall calculate and display overall memorization completion percentage and average assessment scores.

F-028 [Medium] Progress data shall be exportable to PDF format for parent reporting.

3.2.3 Class/Group Management

F-029 [High] Administrators shall be able to create, edit, and delete classes or study groups.

F-030 [High] Administrators shall be able to assign multiple students to a class and assign a primary teacher.

F-031 [Medium] Teachers shall be able to view a list of all students in their assigned classes.

3.3 Daily Memorization Tasks

The Daily Memorization Tasks module enables teachers to assign Quran memorization and revision tasks to students on a daily basis.

3.3.1 Task Creation

F-032 [Critical] Teachers shall be able to create memorization tasks by selecting a Surah and specifying the verse range (Ayah start to Ayah end).

F-033 [Critical] The system shall validate that selected verse ranges exist within the chosen Surah using authenticated Quranic text data.

F-034 [Critical] Teachers shall be able to assign revision tasks separately from new memorization tasks.

F-035 [High] Tasks shall include fields: assignment date, due date, task type (new memorization or revision), Surah, verse range, and optional teacher notes.

F-036 [High] Teachers shall be able to assign tasks to individual students or entire classes in bulk.

F-037 [Medium] The system shall suggest revision tasks based on student's memorization history and spaced repetition algorithm.

3.3.2 Task Delivery

F-038 [Critical] Students shall receive their daily tasks on their personal dashboard organized by due date.

F-039 [Critical] Tasks shall display the full Quranic text (Uthmani script) for the assigned verses with proper Arabic rendering.

F-040 [High] Students shall be able to mark tasks as completed after submitting their memorization recording.

F-041 [High] Overdue tasks shall be visually highlighted on the student dashboard.

F-042 [Medium] The system shall send push notifications for new task assignments and approaching due dates.

3.3.3 Task History

F-043 [High] The system shall maintain a complete task history with status (assigned, submitted, evaluated, overdue).

F-044 [High] Teachers shall be able to view task completion status for all assigned students.

F-045 [Medium] The system shall generate a daily task completion summary for teachers.

3.4 Examinations

The Examinations module provides comprehensive assessment capabilities including weekly and monthly examinations with automated grading for objective question types.

3.4.1 Exam Creation

F-046 [Critical] Teachers shall be able to create examinations with the following attributes: title, description, exam type (weekly/monthly), start date/time, end date/time, time limit, and passing score.

F-047 [Critical] The system shall support three question types: Multiple Choice, True/False, and Short Answer.

F-048 [Critical] Each multiple choice question shall support 2-6 options with exactly one correct answer.

F-049 [High] Teachers shall be able to set point values per question and define the total exam score.

F-050 [High] Teachers shall be able to assign exams to specific students or entire classes.

F-051 [Medium] The system shall support question randomization option for multiple choice questions.

F-052 [Medium] Teachers shall be able to create a question bank for reuse across examinations.

3.4.2 Automated Grading

F-053 [Critical] The system shall automatically grade Multiple Choice and True/False questions upon exam submission.

F-054 [Critical] Short Answer questions shall be flagged for manual grading by the teacher.

F-055 [High] Students shall see their score immediately after submission for fully auto-graded exams.

F-056 [High] The system shall calculate and display: total score, percentage, correct/incorrect count, and pass/fail status.

F-057 [Medium] Partial credit shall be supported for short answer questions during manual grading.

3.4.3 Exam Taking

F-058 [Critical] Students shall only access assigned exams within the defined start and end time window.

F-059 [Critical] The system shall enforce the configured time limit with automatic submission when time expires.

F-060 [High] Students shall be able to navigate between questions and review answers before final submission.

F-061 [High] The exam interface shall display remaining time prominently during the exam session.

F-062 [Medium] The system shall prevent navigation away from the exam page and detect tab switching as potential academic integrity violations.

3.4.4 Exam Results

F-063 [Critical] Teachers shall be able to view all student submissions with individual question responses and scores.

F-064 [High] Students shall be able to view their own exam results with correct answers indicated after the exam period ends.

F-065 [High] The system shall generate a class performance summary showing score distribution and question-level statistics.

F-066 [Medium] Exam results shall be exportable to CSV and PDF formats.

3.5 Notifications System

The Notifications System provides real-time and scheduled communication between the platform and its users across multiple delivery channels.

3.5.1 Announcement Notifications

F-067 [Critical] Administrators shall be able to compose and send announcements to all users, specific roles, or selected classes.

F-068 [Critical] Announcements shall support title, message body, priority level (normal/urgent), and optional attachment.

F-069 [High] Announcements shall be delivered via in-app notification center and push notification (if enabled).

F-070 [High] Users shall be able to view announcement history in their notification center.

3.5.2 Private Notifications

F-071 [Critical] Teachers shall be able to send private notifications to individual students.

F-072 [High] Private notification categories shall include: advice and recommendations, progress reports, warnings, and encouragement messages.

F-073 [High] Students shall receive private notifications in their notification center with read receipts.

F-074 [Medium] Teachers shall be able to schedule notifications for future delivery.

3.5.3 Automated Notifications

F-075 [High] The system shall automatically send notifications for: new task assignments, task due date reminders (24h and 1h before), new exam availability, exam result publication, and new messages.

F-076 [High] Users shall be able to configure their notification preferences per notification type.

F-077 [Medium] Notification preferences shall include: in-app only, push notification, or both.

3.6 Messaging System

The Messaging System enables direct communication between teachers and students for questions, feedback, and task discussions.

3.6.1 Direct Messaging

F-078 [Critical] Students shall be able to send text messages to their assigned teachers.

F-079 [Critical] Teachers shall be able to send text messages to any of their assigned students.

F-080 [High] Messages shall support text content up to 2000 characters.

F-081 [High] Conversations shall be organized as threaded message histories with timestamps.

F-082 [High] Users shall receive real-time message delivery via WebSocket connection.

F-083 [Medium] Users shall be able to search message history by keyword.

3.6.2 Notes and Questions

F-084 [High] Students shall be able to attach notes and questions to specific tasks or exam items.

F-085 [High] Teachers shall receive notifications when students attach questions to tasks.

F-086 [Medium] Teachers shall be able to respond to student notes with threaded replies.

3.7 Memorization Submission and Assessment

The Memorization Submission and Assessment module enables students to record and upload their Quran recitation for teacher evaluation across multiple quality dimensions.

3.7.1 Audio Submission

F-087 [Critical] Students shall be able to record audio directly in the browser or upload pre-recorded audio files.

F-088 [Critical] Supported audio formats shall include: MP3, M4A, and WAV.

F-089 [Critical] Maximum audio file size shall be 20MB per submission.

F-090 [High] The system shall automatically convert uploaded audio to a standardized format (MP3, 128kbps) for storage optimization.

F-091 [High] Students shall be able to review their recording before submission.

F-092 [High] Each submission shall be linked to the corresponding memorization task.

3.7.2 Teacher Assessment

F-093 [Critical] Teachers shall be able to listen to submitted audio recordings through an integrated audio player.

F-094 [Critical] Teachers shall evaluate submissions across three dimensions: Memorization Quality (accuracy of verse recall), Tajweed Performance (application of Tajweed rules), and Fluency (rhythm, pace, and continuity).

F-095 [Critical] Each dimension shall be rated on a defined scale (e.g., 1-5 or 1-10) with optional qualitative feedback.

F-096 [High] Teachers shall be able to provide timestamped comments on specific portions of the recording.

F-097 [High] Assessment results shall be immediately visible to the student upon completion.

F-098 [Medium] The system shall calculate and display a composite assessment score across all three dimensions.

3.7.3 Submission Tracking

F-099 [High] The system shall track all submissions with status: pending review, under review, evaluated, or requires re-submission.

F-100 [High] Students shall be able to view their submission history with associated assessment scores.

F-101 [Medium] Teachers shall be able to filter submissions by student, date range, and review status.

3.8 Tajweed Lessons

The Tajweed Lessons module provides structured instructional content for learning and practicing Quranic recitation rules.

3.8.1 Lesson Content

F-102 [Critical] The system shall provide Tajweed lessons organized by topic: Makharij (articulation points), Sifaat (characteristics of letters), rules of Noon Sakinah, rules of Meem Sakinah, Madd (elongation), and Ghunna.

F-103 [High] Each lesson shall contain: title, description, learning objectives, lesson body content, and practice exercises.

F-104 [High] Lessons shall support rich text content with Arabic script examples.

F-105 [Medium] The system shall track student lesson completion progress.

3.8.2 Multimedia Resources

F-106 [High] Tajweed lessons shall support embedded video content (MP4, up to 100MB per file).

F-107 [High] The system shall support downloadable PDF resources for each lesson.

F-108 [Medium] Audio examples shall be provided demonstrating correct pronunciation for each rule.

F-109 [Medium] Video content shall support streaming with adaptive quality based on connection speed.

3.8.3 Resource Management

F-110 [High] Administrators and Teachers shall be able to upload, edit, and delete Tajweed lesson content.

F-111 [High] Resources shall support categorization by Tajweed topic and difficulty level.

F-112 [Medium] The system shall support content versioning with revision history.

3.9 Live Classes

The Live Classes module provides real-time virtual classroom functionality for interactive memorization sessions, Tajweed instruction, and general educational sessions.

3.9.1 Session Scheduling

F-113 [Critical] Teachers shall be able to schedule live sessions with: title, description, session type (memorization/Tajweed/educational), date, start time, duration, and maximum participants.

F-114 [Critical] Scheduled sessions shall be visible to assigned students on their dashboard and calendar.

F-115 [High] The system shall send notifications to participants 15 minutes before session start.

F-116 [Medium] Teachers shall be able to create recurring sessions (daily/weekly).

3.9.2 Live Session Features

F-117 [Critical] The live session shall support real-time audio and video communication between teacher and students.

F-118 [Critical] The teacher shall have host controls: mute/unmute participants, enable/disable video, share screen, and end session.

F-119 [High] The system shall support text chat alongside audio/video during live sessions.

F-120 [High] Teachers shall be able to share their screen to display Quranic text or educational materials.

F-121 [Medium] Sessions shall support a virtual hand-raise feature for student questions.

F-122 [Medium] The system shall support recording of live sessions for later review (stored for 30 days).

3.9.3 Session Management

F-123 [High] Administrators shall be able to view all scheduled and completed sessions.

F-124 [High] The system shall track attendance with join/leave timestamps for each participant.

F-125 [Medium] Teachers shall be able to generate attendance reports per session.

3.10 Islamic Content Center

The Islamic Content Center provides supplementary educational resources to enrich students' Islamic knowledge beyond Quran memorization and Tajweed.

3.10.1 Daily Supplications

F-126 [High] The system shall provide a collection of daily supplications (Duas) organized by occasion: Morning (Adhkar al-Sabah), Evening (Adhkar al-Masa), Before Sleep, and Travel.

F-127 [High] Each supplication shall include: Arabic text, transliteration, translation, and audio recitation.

F-128 [Medium] Students shall be able to mark supplications as memorized and track completion progress.

3.10.2 Seerah Lessons

F-129 [High] The system shall provide structured lessons on the Biography of Prophet Muhammad (Seerah) covering: Birth and Early Life, Prophethood and Early Revelation, Migration to Medina (Hijrah), Major Battles and Events, and The Farewell Pilgrimage.

F-130 [High] Seerah lessons shall support rich text, images, and video content.

F-131 [Medium] The system shall track lesson completion progress for each student.

3.10.3 Stories of Companions

F-132 [High] The system shall provide biographical content for the Four Rightly Guided Caliphs: Abu Bakr al-Siddiq, Umar ibn al-Khattab, Uthman ibn Affan, and Ali ibn Abi Talib.

F-133 [High] The system shall provide biographical content for notable female companions: Khadijah bint Khuwaylid, Aisha bint Abi Bakr, Asma bint Abi Bakr, and Nusaybah bint Ka'ab.

F-134 [Medium] Companion stories shall include structured sections: early life, role in Islam, key contributions, and lessons learned.

3.10.4 Content Management

F-135 [High] Administrators shall be able to create, edit, and organize all Islamic content.

F-136 [High] Content shall support rich text editing with Arabic script support.

F-137 [Medium] Content items shall support categorization, tagging, and searchability.

3.11 Reports and Analytics

The Reports and Analytics module provides data-driven insights into student performance, teacher effectiveness, and platform utilization.

3.11.1 Student Performance Reports

F-138 [Critical] Teachers shall be able to generate individual student reports showing: memorization progress over time, examination score trends, task completion rates, and assessment score averages.

F-139 [High] Reports shall support date range filtering (weekly, monthly, custom range).

F-140 [High] Reports shall be exportable in PDF format.

F-141 [Medium] The system shall automatically highlight students with declining performance or missed assignments.

3.11.2 Class Performance Reports

F-142 [High] Teachers shall be able to generate class-level reports with aggregate statistics: average scores, completion rates, and performance distribution.

F-143 [Medium] Class reports shall include comparative analysis across students.

3.11.3 Administrative Reports

F-144 [High] Administrators shall be able to view platform-wide statistics: total active users, total memorization tasks completed, examination participation rates, and live class utilization.

F-145 [High] Administrators shall be able to view teacher activity reports showing tasks assigned, exams created, and sessions conducted.

F-146 [Medium] The system shall generate monthly automated summary reports delivered to administrators.

3.12 Content Management

The Content Management module provides administrative tools for managing all educational content across the platform.

3.12.1 Quranic Text Management

F-147 [Critical] The system shall maintain authenticated Quranic text in Uthmani script with proper verse numbering.

F-148 [High] Quranic text shall be searchable by Surah name, verse number, and keyword (Arabic).

F-149 [High] The system shall support multiple verified translations for display alongside Arabic text.

3.12.2 Media Asset Management

F-150 [High] Administrators shall be able to upload, organize, and delete media files (audio, video, PDF).

F-151 [High] The system shall generate thumbnails for video content.

F-152 [Medium] Media files shall support metadata: title, description, upload date, file size, and usage count.

3.13 Statistics Dashboard

The Statistics Dashboard provides at-a-glance visual summaries of key performance indicators for administrators and teachers.

3.13.1 Dashboard Widgets

F-153 [High] The Administrator dashboard shall display: total students, total teachers, active classes, tasks completed this week, exams conducted this month, and upcoming live sessions.

F-154 [High] The Teacher dashboard shall display: assigned student count, pending task submissions, pending exam grading, upcoming scheduled sessions, and recent student activity.

F-155 [High] The Student dashboard shall display: today's tasks, upcoming exams, recent assessment scores, unread notifications, and live class schedule.

3.13.2 Data Visualization

F-156 [Medium] The dashboard shall include charts: memorization progress line chart, exam score bar chart, task completion pie chart, and weekly activity heatmap.

F-157 [Medium] All dashboard data shall update in real-time without requiring page refresh.

4. External Interface Requirements

4.1 User Interfaces

The system shall provide three distinct user interface experiences optimized for each user role. All interfaces shall implement responsive design to support desktop, tablet, and mobile web browsers.

4.1.1 Administrator Interface

F-158 [Critical] The Administrator interface shall provide a comprehensive dashboard with navigation to: User Management, Content Management, Live Classes Management, Exam Management, Reports, Announcements, and System Settings.

F-159 [High] User management screens shall include bulk operation capabilities for student enrollment and teacher assignment.

F-160 [High] All management screens shall include search, filter, and pagination for large datasets.

4.1.2 Teacher Interface

F-161 [Critical] The Teacher interface shall provide quick-access widgets for: student list, task creation, exam creation, live session scheduling, and pending submissions.

F-162 [Critical] The task assignment interface shall include a Surah/verse selector with autocomplete and Quranic text preview.

F-163 [High] The audio review interface shall include an integrated audio player with playback controls, timestamp annotation, and scoring input fields.

F-164 [High] The exam creation interface shall support drag-and-drop question reordering and live preview.

4.1.3 Student Interface

F-165 [Critical] The Student interface shall present a personalized dashboard with today's tasks prominently displayed.

F-166 [Critical] The memorization task view shall display Quranic text with proper Arabic rendering and audio recording controls.

F-167 [High] The exam-taking interface shall present one question at a time with clear navigation controls and a visible countdown timer.

F-168 [High] All student interfaces shall support RTL (Right-to-Left) layout for Arabic text display.

4.1.4 Cross-Cutting UI Requirements

F-169 [Critical] All interfaces shall support Arabic language display with proper RTL text direction.

F-170 [Critical] The system shall be usable on screen sizes from 320px (mobile) to 1920px (desktop).

F-171 [High] Page load time shall not exceed 3 seconds on a 4G connection.

F-172 [High] All user actions that trigger server requests shall display loading indicators.

F-173 [Medium] The interface shall support dark mode toggle for reduced eye strain during evening study sessions.

4.2 Hardware Interfaces

The system shall interface with the following hardware components through standard web APIs:

- ☐ Microphone: Required for student memorization recording and teacher live class audio
- ☐ Camera: Required for teacher live class video and optional for student video calls
- ☐ Speakers/Headphones: Required for audio playback of Tajweed examples and live sessions
- ☐ Storage: Browser local storage for offline task caching and draft saving

4.3 Software Interfaces

System	Interface Type	Purpose
Firebase Cloud Messaging	REST API	Push notification delivery to web and mobile clients
Cloud Object Storage	S3-compatible API	Media file storage for audio recordings and videos
Daily.co / Twilio	REST API + SDK	Live video session management and WebRTC signaling
SMTP Server	SMTP	Transactional email for password reset and notifications
PostgreSQL	TCP/IP + SQL	Primary data persistence and query processing

4.4 Communications Interfaces

The system shall use the following communication protocols and standards:

- ☐ HTTPS/TLS 1.3: All client-server communication shall be encrypted
- ☐ WebSocket: Real-time messaging and live class signaling

- ☐ WebRTC: Peer-to-peer audio/video streaming for live classes
- ☐ SMTP: Email delivery for password reset and notifications
- ☐ REST API: JSON-based API with OpenAPI 3.0 documentation
- ☐ FCM HTTP v1 API: Push notification delivery to mobile and web clients

5. Nonfunctional Requirements

This section specifies the quality attributes, performance criteria, and system constraints that the Quran Learning Platform must satisfy. Each nonfunctional requirement is identified using the format [NF-XXX].

5.1 Performance Requirements

NF-001 [Critical] The system shall support a minimum of 1000 concurrent users without performance degradation.

NF-002 [Critical] API response time for standard CRUD operations shall not exceed 500ms under normal load (P95).

NF-003 [Critical] Page initial load time shall not exceed 3 seconds on a 10 Mbps connection.

NF-004 [High] Audio file upload shall complete within 30 seconds for a 10MB file on a 5 Mbps upload connection.

NF-005 [High] Video streaming shall start within 5 seconds and maintain a buffer of at least 5 seconds.

NF-006 [High] Live class latency shall not exceed 500ms for audio and 150ms for video (teacher to student).

NF-007 [High] Dashboard data queries shall complete within 2 seconds for date ranges up to 90 days.

NF-008 [Medium] Search queries across Quranic text shall return results within 1 second.

NF-009 [Medium] The system shall handle examination submission spikes of up to 200 concurrent submissions without data loss.

5.2 Security Requirements

NF-010 [Critical] All passwords shall be hashed using bcrypt (cost factor ≥ 12); plaintext password storage is prohibited.

NF-011 [Critical] JWT tokens shall be signed using RS256 with 2048-bit keys; token secrets shall be rotated every 90 days.

NF-012 [Critical] All API endpoints shall validate authentication tokens on every request except public endpoints.

NF-013 [Critical] The system shall enforce HTTPS for all communications; HTTP requests shall be permanently redirected to HTTPS.

NF-014 [Critical] The system shall implement SQL injection protection through parameterized queries (Drizzle ORM parameterized queries).

NF-015 [Critical] The system shall implement XSS protection through output encoding and Content Security Policy headers.

NF-016 [High] All user actions modifying data shall be logged with timestamp, user ID, action type, and affected resource.

NF-017 [High] The system shall enforce rate limiting: 100 requests per minute per authenticated user, 20 requests per minute per unauthenticated IP.

NF-018 [High] Audio recordings and personal data shall be encrypted at rest using AES-256.

NF-019 [High] Session tokens shall be invalidated upon logout and password change.

NF-020 [Medium] The system shall implement brute-force protection with progressive delays and account lockout after 10 failed attempts.

NF-021 [Medium] Security headers shall include: Strict-Transport-Security, X-Content-Type-Options, X-Frame-Options, and Referrer-Policy.

5.3 Availability and Reliability

NF-022 [Critical] The system shall maintain 99.9% uptime during scheduled operational hours (measured monthly).

NF-023 [Critical] Database backups shall be performed automatically every 24 hours with 30-day retention.

NF-024 [High] The system shall support zero-downtime deployments using rolling update strategy.

NF-025 [High] In the event of service degradation, core functions (authentication, task viewing, exam taking) shall remain available.

NF-026 [Medium] The system shall implement health check endpoints for monitoring system component status.

NF-027 [Medium] Mean Time To Recovery (MTTR) shall not exceed 1 hour for critical system failures.

5.4 Maintainability

NF-028 [Critical] The codebase shall follow established coding standards (ESLint for JavaScript/TypeScript, consistent naming conventions).

NF-029 [Critical] Database schema changes shall be managed through versioned migration files using Drizzle migration system.

NF-030 [High] API documentation shall be maintained using OpenAPI 3.0 specification and be auto-generated from code annotations.

NF-031 [High] The system shall implement structured logging with correlation IDs for request tracing across services.

NF-032 [Medium] Component-level unit test coverage shall be minimum 70% for business logic modules.

NF-033 [Medium] The system shall use containerized deployment (Docker) for consistent environments across development, staging, and production.

5.5 Usability

NF-034 [Critical] The interface shall be intuitive requiring no training for students; teacher features shall include guided onboarding.

NF-035 [High] All error messages shall be user-friendly and provide actionable guidance (no raw error codes or stack traces).

NF-036 [High] Form validation shall provide immediate feedback without requiring form submission.

NF-037 [High] Arabic text shall render correctly with appropriate font selection and RTL layout support.

NF-038 [Medium] The system shall be WCAG 2.1 AA compliant for accessibility, including screen reader support and keyboard navigation.

NF-039 [Medium] Color contrast ratios shall meet WCAG AA standards (minimum 4.5:1 for normal text, 3:1 for large text).

5.6 Scalability

NF-040 [High] The system architecture shall support horizontal scaling of API servers to handle 10x user growth.

NF-041 [High] Database queries shall be optimized with appropriate indexing; query execution plans shall be reviewed quarterly.

NF-042 [Medium] Static content (images, videos, PDFs) shall be served through CDN for reduced server load.

NF-043 [Medium] The system shall implement database connection pooling with a minimum of 10 and maximum of 100 connections.

5.7 Internationalization and Localization

NF-044 [Critical] The system shall support Arabic and English languages with all user-facing text externalized in resource files.

NF-045 [Critical] Arabic interface shall use RTL (Right-to-Left) layout; English interface shall use LTR layout.

NF-046 [High] Quranic text shall be displayed in authenticated Uthmani script with optional transliteration.

NF-047 [Medium] Date and time shall be displayed in the user's local timezone with Hijri date option.

NF-048 [Low] The architecture shall support additional language additions without code changes.

5.8 Compliance

NF-049 [Critical] The system shall comply with GDPR requirements for EU users including: data subject access requests, right to erasure, and data portability.

NF-050 [Critical] Personal data collection shall require explicit consent with clear privacy policy disclosure.

NF-051 [High] The system shall maintain audit logs for all personal data access and modifications for compliance reporting.

NF-052 [High] Children under 13 shall require parental consent for account creation.

NF-053 [Medium] The system shall implement data retention policies with automatic purging of inactive accounts after 2 years.

6. Data Requirements

6.1 Data Models

The system shall maintain the following core data entities with their specified attributes:

Entity	Key Attributes
User	id, email, password_hash, full_name, role (enum), phone, avatar_url, is_active, created_at, updated_at
StudentProfile	id, user_id, enrollment_date, teacher_id, class_id, current_juz, current_surah, notes
Class	id, name, description, teacher_id, created_by, created_at
MemorizationTask	id, teacher_id, student_id, task_type (new/revision), surah_number, ayah_start, ayah_end, notes, assigned_date, due_date, status
MemorizationSubmission	id, task_id, student_id, audio_url, submitted_at, status, teacher_notes
Assessment	id, submission_id, memorization_score, tajweed_score, fluency_score, overall_score, feedback, assessed_at
Exam	id, title, description, exam_type, teacher_id, start_time, end_time, time_limit_minutes, passing_score, status
Question	id, exam_id, question_type, question_text, options (JSON), correct_answer, points, order_index
ExamSubmission	id, exam_id, student_id, started_at, submitted_at, total_score, status
Answer	id, submission_id, question_id, student_answer, is_correct, points_awarded
LiveSession	id, title, description, session_type, teacher_id, scheduled_at, duration_minutes, max_participants, meeting_url, status
Attendance	id, session_id, user_id, joined_at, left_at
Notification	id, sender_id, recipient_id, type, title, message, is_read, created_at
Message	id, sender_id, recipient_id, content, created_at, is_read
TajweedLesson	id, title, description, category, content, video_url, pdf_url, order_index, created_by
IslamicContent	id, title, content, category (supplication/seerah/companion), subcategory, media_urls, order_index

6.2 Data Storage

NF-054 [Critical] All structured data shall be stored in PostgreSQL 15+ with UTF-8 encoding for full Unicode support.

NF-055 [Critical] Media files (audio recordings, videos, PDFs, profile photos) shall be stored in cloud object storage with CDN distribution.

NF-056 [High] Database schema shall be managed through Drizzle ORM migration files with version control.

NF-057 [High] File storage shall support automatic virus scanning on upload.

NF-058 [Medium] Media files shall be automatically transcoded to optimized formats (audio: MP3 128kbps, video: H.264 720p).

6.3 Data Retention

NF-059 [High] User account data shall be retained for the duration of active enrollment plus 2 years.

NF-060 [High] Audio memorization submissions shall be retained for 1 year after submission date.

NF-061 [High] Examination data and results shall be retained for 5 years.

NF-062 [High] System audit logs shall be retained for 1 year.

NF-063 [Medium] Live class recordings shall be retained for 30 days after the session.

NF-064 [Medium] Deleted user accounts shall be soft-deleted with data anonymized after the retention period expires.

7. Minimum Viable Product Phasing

The development and release of the Quran Learning Platform shall follow a phased approach to deliver value incrementally while managing risk and gathering early user feedback.

7.1 Phase 1: Core Platform (Weeks 1-8)

Phase 1 establishes the foundational platform with essential user management, task assignment, examination, and communication features.

Req. IDs	Feature	Description
F-001 to F-005	User Registration	Admin-managed account creation
F-006 to F-018	Authentication	Login, JWT, RBAC, password management
F-019 to F-031	Student Management	Profiles, progress monitoring, classes
F-032 to F-045	Daily Memorization Tasks	Task creation, delivery, history
F-046 to F-066	Examinations	Exam creation, auto-grading, results
F-067 to F-077	Notifications	Announcements, automated alerts
F-078 to F-086	Messaging System	Direct messaging, notes/questions
NF-001 to NF-021	Security	Auth, encryption, rate limiting

7.2 Phase 2: Assessment and Live Instruction (Weeks 9-14)

Phase 2 introduces audio-based memorization assessment, live virtual classes, and enhanced progress tracking capabilities.

Req. IDs	Feature	Description
F-087 to F-101	Memorization Assessment	Audio submission, teacher review, scoring
F-113 to F-125	Live Classes	Scheduling, WebRTC sessions, attendance
F-024 to F-028	Progress Tracking	Charts, reports, analytics
F-153 to F-157	Dashboard	Role-based dashboards, visualizations

7.3 Phase 3: Content and Mobile (Weeks 15-22)

Phase 3 delivers the Islamic Content Center, administrative analytics, and the mobile application for iOS and Android platforms.

Req. IDs	Feature	Description
F-102 to F-112	Tajweed Lessons	Lesson content, multimedia, management
F-126 to F-137	Islamic Content Center	Supplications, Seerah, companion stories
F-138 to F-146	Reports and Analytics	Performance reports, admin summaries
F-147 to F-152	Content Management	Quranic text, media assets
Mobile App	React Native	iOS and Android application

7.4 Phase Dependencies

Phase 2 depends on the completion of Phase 1 core authentication and student management features. Phase 3 depends on Phase 2 assessment features for progress tracking integration. The mobile application development can proceed in parallel with Phase 3 backend features but requires Phase 1 API completion for core functionality.

Appendix A: Recommended Technology Stack

The following technology stack is recommended for the implementation of the Quran Learning Platform. This selection prioritizes developer productivity, ecosystem maturity, and alignment with the specified technical constraints.

A.1 Backend

Component	Technology	Rationale
Runtime	Node.js 20 LTS	Long-term support, modern JavaScript features, high performance for I/O-bound operations
Framework	Express.js	Mature, minimal, extensive middleware ecosystem
Database	PostgreSQL 15+	ACID compliance, full Unicode support, JSON columns, excellent reliability
ORM	Drizzle ORM	Type-safe, SQL-like query builder, excellent migration support
Authentication	JWT (jsonwebtoken)	Stateless, scalable, industry-standard token-based auth
Validation	Zod	TypeScript-first schema validation with excellent DX
API Documentation	Swagger/OpenAPI	Auto-generated from Zod schemas, interactive documentation

A.2 Frontend

Component	Technology	Rationale
Framework	React 18	Component-based, virtual DOM, extensive ecosystem
Language	TypeScript	Type safety, improved IDE support, reduced runtime errors
Styling	Tailwind CSS	Utility-first, rapid development, consistent design system
State Management	Zustand	Lightweight, minimal boilerplate, excellent TypeScript support

Component	Technology	Rationale
Data Fetching	TanStack Query	Server state management, caching, background updates
Routing	React Router v6	Declarative routing, code splitting support
UI Components	shadcn/ui	Accessible, customizable, Tailwind-compatible components

A.3 Mobile Application

Component	Technology	Rationale
Framework	React Native	Shared codebase with web, native performance, large community
Navigation	React Navigation	Native navigation patterns, deep linking support
State Management	Zustand	Shared logic with web frontend

A.4 Infrastructure

Component	Technology	Rationale
Hosting	Render or Railway	Managed platform, automatic deployments, PostgreSQL included
Media Storage	Cloudflare R2 or AWS S3	Object storage for audio/video/files
CDN	Cloudflare	Global edge caching, DDoS protection, free tier
Push Notifications	Firebase Cloud Messaging	Cross-platform push, reliable delivery
Real-time	Socket.io	WebSocket abstraction, fallback support, room management
Live Video	Daily.co or Twilio	Managed WebRTC infrastructure, low latency

A.5 Development Tools

Component	Technology	Rationale
Version Control	Git + GitHub	Branch protection, code review, CI/CD integration
CI/CD	GitHub Actions	Automated testing, linting, deployment
Testing	Vitest + Playwright	Unit testing, E2E testing, visual regression
Linting	ESLint + Prettier	Code quality, consistent formatting
Monitoring	Sentry + LogRocket	Error tracking, session replay



Quran Learning Platform

Software Requirements Specification

For inquiries and project information:

platform@quranlearning.edu | www.quranlearning.edu

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